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Abstract

The hardness and brittleness of transparent ceramic make it difficult to be prepared in large scale and limit its application. This paper tries to solve this problem from the perspective of realizing its own reliable joining. Without interlayer and preheating, MgAl₂O₄ transparent ceramics have been directly welding by ultrashort pulse laser for the first time. The typical structure of the joint was systematically characterised, it can be found that at the bonding area, parts of MgAl₂O₄ crystals were transformed into amorphous. Under the optimal welding parameters (laser power: 550mW, scanning speed: 0.1 mm/s), the joint achieved a maximum shear strength of approximately 28MPa. This paper provides a new methodology for MgAl₂O₄ transparent ceramics joining,

Outline

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owning high efficiency and simplicity, which makes efforts for broadening the application of transparent ceramics.

Keywords

Ultrashort pulse laser; MgAl₂O₄ ceramic; Microstructure; Mechanical property

Acknowledgments

Data availability

References

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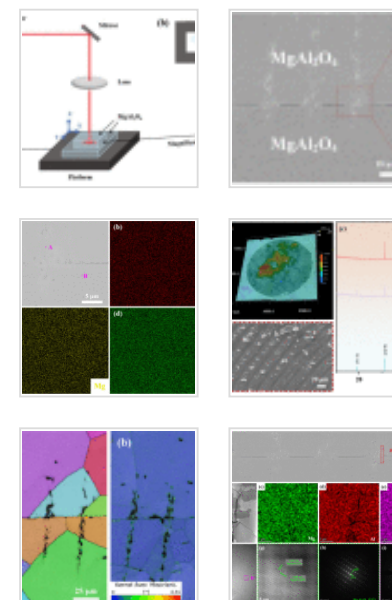
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1. Introduction

In recent years, transparent ceramic materials have been widely used in multiple dimensions of production and life. Their transparency provides them with the possibility of being an optical window, in the meantime, the advantages of high strength, high temperature resistance and abrasive performance make them more superior to traditional glass materials in many fields [[1], [2], [3]]. For instance, MgAl₂O₄ transparent ceramic not only has the high strength and impact resistance of ordinary ceramics, but also has good optical properties of good transmittance from ultraviolet to near infrared band [4]. For this reason, MgAl₂O₄ ceramic is often used in prism, optical window and other fields [5,6]. However, the preparation size of MgAl₂O₄ transparent ceramic is limited [7], two main challenges it faced with are: i) Transparent MgAl₂O₄ ceramics are hard to be sintered in large scales. This is because ceramics are brittle materials and the cooling rate could be greatly different in the middle and at the edge of the large ceramics. The temperature gradient could lead to high-level stress resulting in the breaking up of the large ceramics. ii) It could be difficult to process transparent ceramic into complex shapes due to the high hardness of the material. Therefore, the reliable joining between MgAl₂O₄ transparent ceramics is propitious to their wide application.

Figures (13)



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Tables (1)

Table 1. The EDS analysis results of the v...

One practical approach to overcome the above two mentioned challenges is by joining small pieces of ceramics into larger component or bonding simple ceramic parts into complex shapes. At present, there are two main joining methods for MgAl₂O₄ transparent ceramics, which are brazing [8], [9], [10] and flash joining [11,12]. However, the brazing process is time-consuming and power-wasting [13]. Besides, the indispensable interlayer in the process of brazing also leads to residual stress arising from the thermal expansion coefficient (CTE) mismatch between the ceramic and the interlayer, which makes the joint less reliable. Flash joining, although has played a great advantage in saving time and efficiency, but the joining parameters can be affected by the size and shape of the sample, which is not furnished with universality. Therefore, it possesses great urgency to develop a new way to realize the joining of MgAl₂O₄ transparent ceramics.

The emergence of ultrashort pulse laser welding technology provides a new thread for the joining of transparent materials. Ultrashort pulse laser refers to the laser whose pulse width is less than picosecond (10^{-12} s). By focusing the USPL, the peak optical power density in the focal field region may exceed 10^{12} W/cm², which can induce a series of nonlinear effects and local phase transitions of transparent materials [14]. Therefore, ultrashort pulsed laser is widely used in cutting, surface treatment, welding and so on. For example, Ding et al. [15] used femtosecond laser optical field modulation combined with wet etching to make bionic micropores on the surface of ZnS glass, and the treated ZnS glass showed excellent optical properties. Inspired by moth's eyes, Gao et al. [16] used second harmonic femtosecond laser to directly scan and assist chemical etching on the surface of silica glass, and prepared periodic arrays of nano-cones or nano-holes. The results show that the average transmittance of the treated silica glass is more than 99.5% in a wide wavelength range, which is much higher than that of the untreated sample. And the reason for the performance improvement is that the tapered structure prepared by femtosecond laser creates a refractive index gradient layer from air to silica glass, which effectively eliminates the interface reflection.

For ultrashort pulse laser welding, it has the following advantages. (i) The pulse duration of ultrashort pulse laser is extremely short, which is much shorter than the relaxation

time of excited electrons in the material. This characteristic can limit the heat accumulation to a local area, avoiding heating the whole component, which can weaken the influence caused by thermal stress. (ii) Ultrashort pulse laser welding has three-dimensional spatial selectivity, which provides theoretical possibility for preparing complex structural components. (iii) Ultra-short pulse laser welding is time-saving and environment-friendly. This improves the production efficiency to a certain extent. T. Tamaki et al. [17], who is the first one to use ultrashort pulse laser to realize the joining between two pieces of silica glass. Since then, the door of ultrashort pulse laser welding (USPL) transparent materials has been opened. A lot of research shows that femtosecond laser welding is also suitable for a variety of glasses [[18], [19], [20]], polymers [[21], [22], [23]] and their joining with metals and so on. This undoubtedly broadens the application range of femtosecond laser welding. In our previous research [24], femtosecond laser welding between transparent ceramics and metals was reported. But for transparent ceramic itself, a hard and brittle material, its femtosecond laser welding is rarely reported. Therefore, it possesses great necessity to focus on the joining between transparent ceramics.

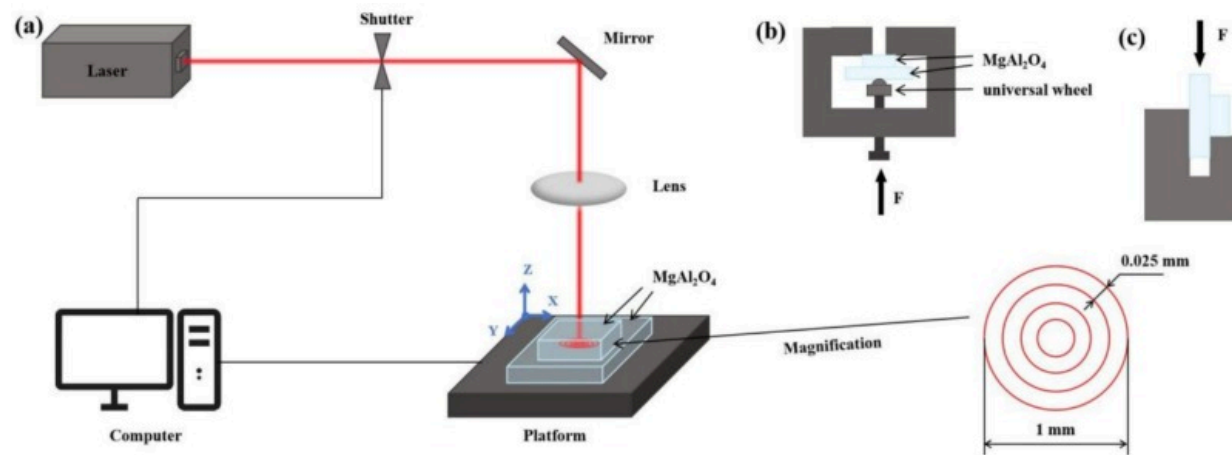
In this paper, MgAl₂O₄ transparent ceramic has been chosen as the based material for ultrashort pulse laser welding. The feasibility of USPL has been confirmed and the typical microstructure was thoroughly characterized by SEM, TEM, EBSD and XRD. The influence of laser power and scanning speed on the microstructure and mechanical properties of the joint has been elucidated detailedly. This paper aims to provide a new direction and reference for the field of ultrashort pulse laser welding transparent ceramics.

2. Experimental

2.1. Materials and welding processing

Ultrashort pulse laser welding equipment was used in this study, equipped with the laser source which can emit femtosecond laser with the wavelength of 800nm, the repetition rate of 1 kHz, the pulse width less than 100fs and the spot diameter of ~15μm at the focal

plane. The processing platform for the experiment supports the movement in X, Y and Z directions, at the same time, can rotate around X and Z axes. The focus position is determined by a CCD camera coaxial with the laser beam, whose phase plane is adjusted to coincide with the focal plane of the laser. During the welding process, laser scanning path is series of concentric circles with a diameter of 1 mm and a hatching distance of 0.025 mm, which is shown in Fig. 1a.



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Fig. 1. Schematic diagrams of (a) USPL welding process, (b) fixture, (c) shear strength test.

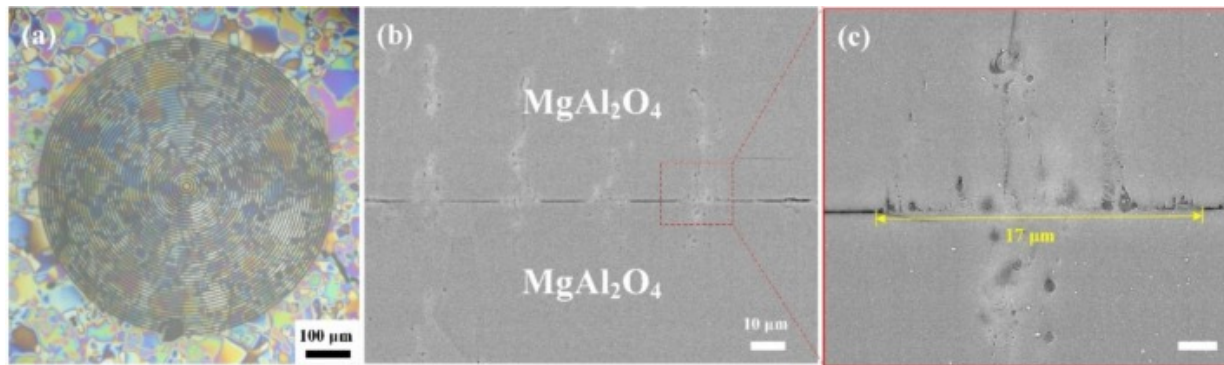
Transparent MgAl_2O_4 ceramics with different sizes were used in our experiment, which were purchased from Hefei Crystal Technical Material Co., Ltd. One of them was polished on both sides with the size of 5 mm×5 mm×3 mm, while another was polished on one side with the size of 8 mm×6 mm×3 mm. The surface roughness of the polished MgAl_2O_4 ceramics is less than 10 nm. Before welding, two pieces of MgAl_2O_4 ceramics with different sizes were ultrasonic washed in alcohol for 10 min and then fastened together with a fixture. The standard to meet the welding requirements is that Newton's rings with light and dark stripes can be seen clearly at the interface, which can ensure the successful joining.

2.2. Characterization

The microstructure of the joint was characterized by Scanning Electron Microscopy (SEM, Zeiss MERLIN Compact, Germany) with an equipped EDS. TEM (Talos F200x, FEI, USA) was used to further characterize the phase composition in the joint. In addition, the phases in the joint were identified by XRD (D8 Advanced, Bruker, Germany). The fracture morphology was observed by ultra-depth optical microscope (VHX-1000E, KEYENCE, Japan). For the purpose of analyzing the grain size and orientation information, electron backscattered diffraction (EBSD) was utilized in this study. The shear strengths of the joints were tested by electronic universal testing machine (AG-X Plus, Instron, USA) with the loading rate of 0.5 mm/s and used the fixture shown in [Fig. 1b](#). Each measurement tests at least 3 specimens.

3. Results and discussion

The typical macro-morphology and microstructure of the joint obtained 550mW/0.1 mm/s are shown in [Fig. 2](#). As shown in [Fig. 2a](#), there is no obvious crack in the welding area, and Newton's rings at the interface are clearly visible, which reflects that MgAl₂O₄ ceramics have achieved good bonding. It can be seen in [Fig. 2b](#) that the crack-free MgAl₂O₄/MgAl₂O₄ joint with some filamentary laser affected zones with the length about 50 μm above and below the interface. The contrast between the laser affected zones and the base material was slightly different. When enlarged one of the welding zones, it can be found that there are some microvoids in the laser affected zones, which are slightly less dense than the base materials with the bonding width about 17 μm.

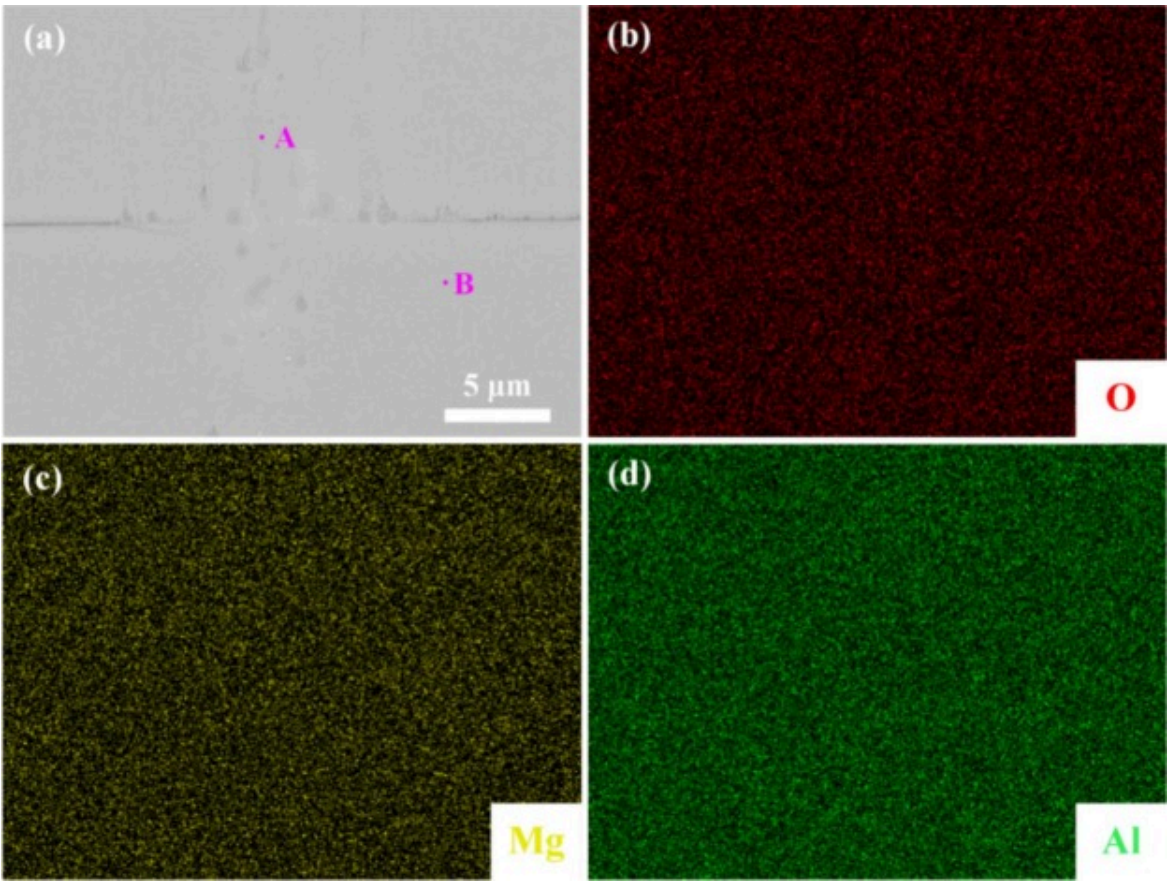


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Fig. 2. (a) Macro-morphology of $\text{MgAl}_2\text{O}_4/\text{MgAl}_2\text{O}_4$ joint; (b) The SEM image of the $\text{MgAl}_2\text{O}_4/\text{MgAl}_2\text{O}_4$ joint, (b₁) Magnification of the region in (b).

Fig. 3 shows the mapping results near the bonding area, which aims to determine the element distribution in the joint. There is no obvious spatial gradient change of O, Mg and Al in the welding zone, which indicates that there is no element volatilization during the welding process. The light phase A in the welding zone and the dark phase B in the base material are analyzed by EDS and the results are summarized in Table 1. It can be found that the light phase A in the welding zone is the same as dark phase B in the base material.



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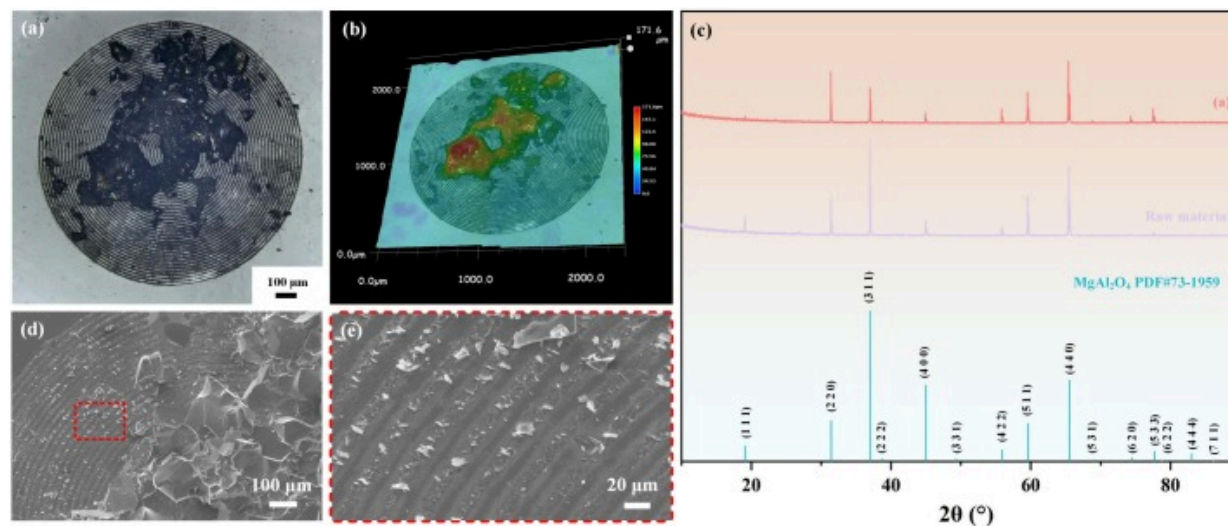
Fig. 3. (a) BSE image of MgAl₂O₄/MgAl₂O₄ joint, the element distribution of (b) O, (c) Mg, (d) Al.

Table 1. The EDS analysis results of the various phases in the joint shown in Fig. 3a (at. %).

Position	O	Mg	Al	Possible phase
A	52.86	14.49	32.65	MgAl ₂ O ₄

Position	O	Mg	Al	Possible phase
B	53.26	13.67	33.07	MgAl ₂ O ₄

The fracture morphology of the joint was observed and then analyzed by XRD. As shown in Fig. 4a–b, the fracture mainly occurred near the interface between MgAl₂O₄ ceramics with some different-size MgAl₂O₄ crystal fragments remained on the surface of one ceramic substrate. The SEM image of fracture surface (Fig. 4d) indicates that the remaining MgAl₂O₄ crystal shows obvious brittle fracture characteristics. It can be found that there are a lot of small pieces of MgAl₂O₄ above the welding line by enlarging the area near the welding line, which indicates that the ceramic in the action area may be melted and resolidified by femtosecond laser to realize the joining between MgAl₂O₄ ceramics [25]. However, the resolidified ceramics are less dense than the base materials, eventuating the uneven-featured fracture. The result of XRD analysis shown in Fig. 4c further confirms that no new phase was formed in the joint.

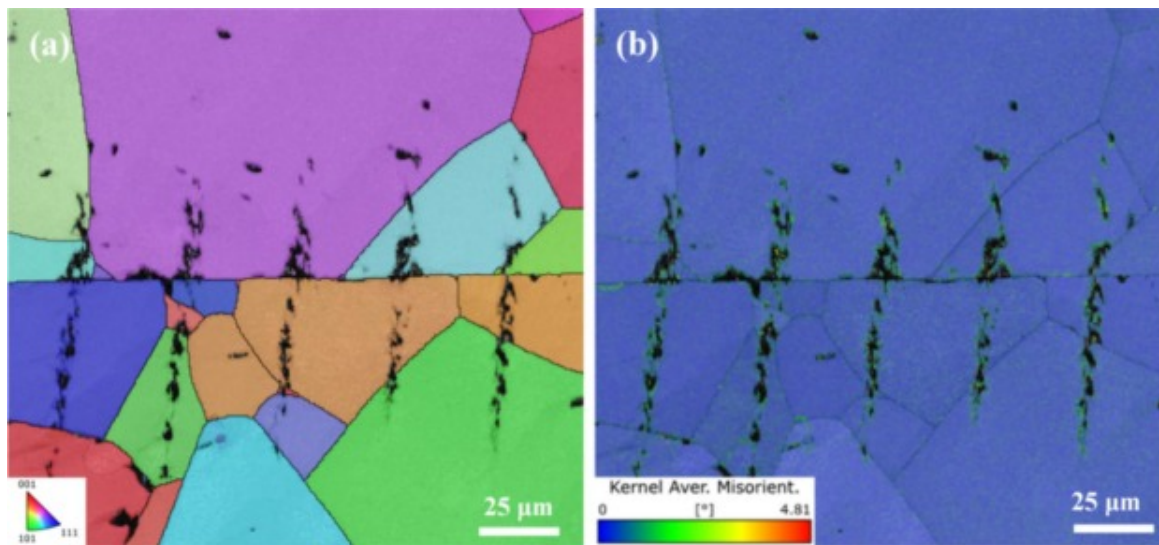


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Fig. 4. Fracture morphology of $\text{MgAl}_2\text{O}_4/\text{MgAl}_2\text{O}_4$ joint: (a) macroscopic image, (b) three-dimensional image, (c) XRD result, (d) SEM image of fracture, and (e) enlarged zone of (d).

For the purpose of analyzing the grain change in MgAl_2O_4 ceramics after femtosecond laser irradiation, EBSD is carried out near the interface with the result shown in Fig. 5. As shown in Fig. 5a, no grain coarsening phenomenon of gradient transformation was detected in the region far from the interface. It is worth noting that there is a filamentary black area along the laser incident direction representing dead pixels, which corresponds to the light filamentary laser action area in Fig. 2b. It is judged that this area may be amorphous, so it was not identified in EBSD test. According to the KAM (Kernel Amplitude Matrix) in Fig. 5b, it can be discerned that the KAM values are all high near the laser affected area indicating the gathering of dislocations adjacent to here [26], which might be due to the femtosecond laser processing with high peak power density.

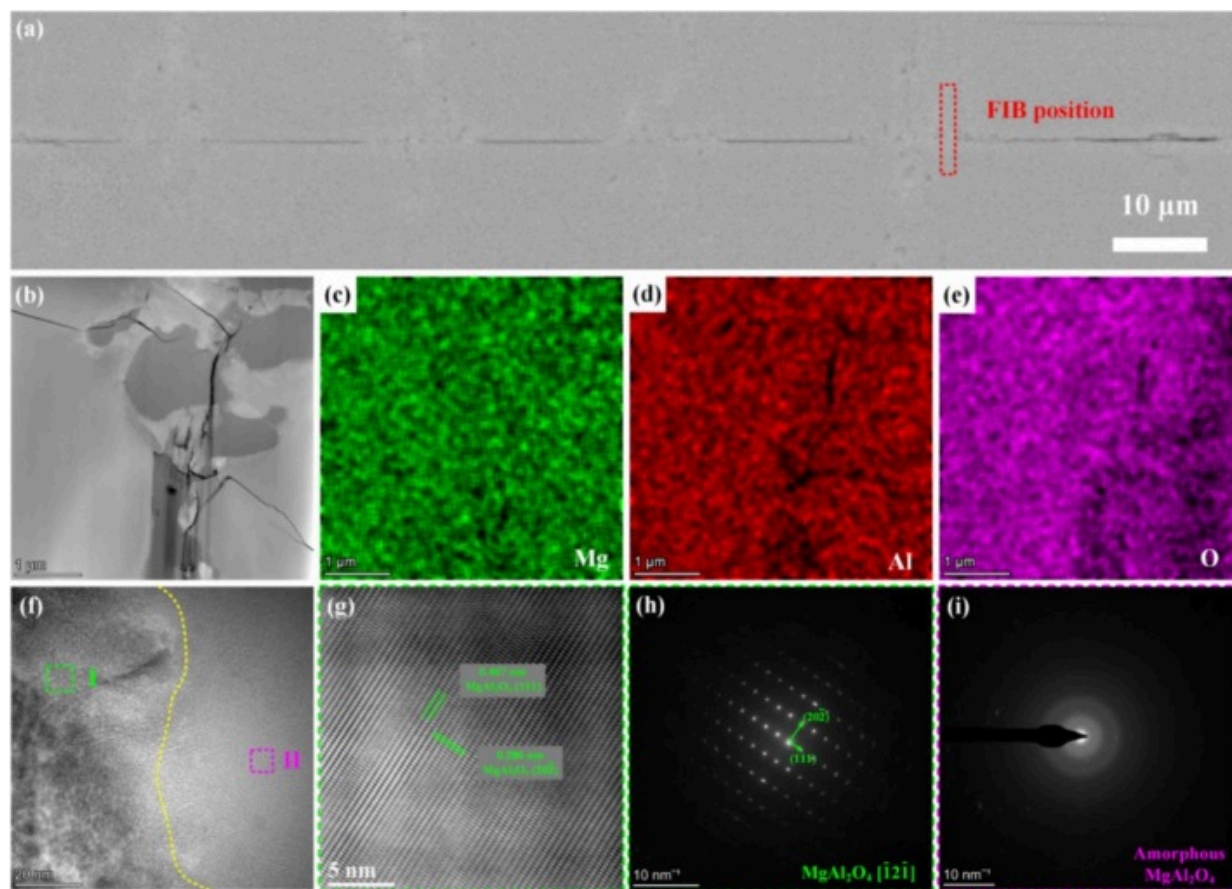


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Fig. 5. Diagrams of (a) EBSD analysis, (b) KAM.

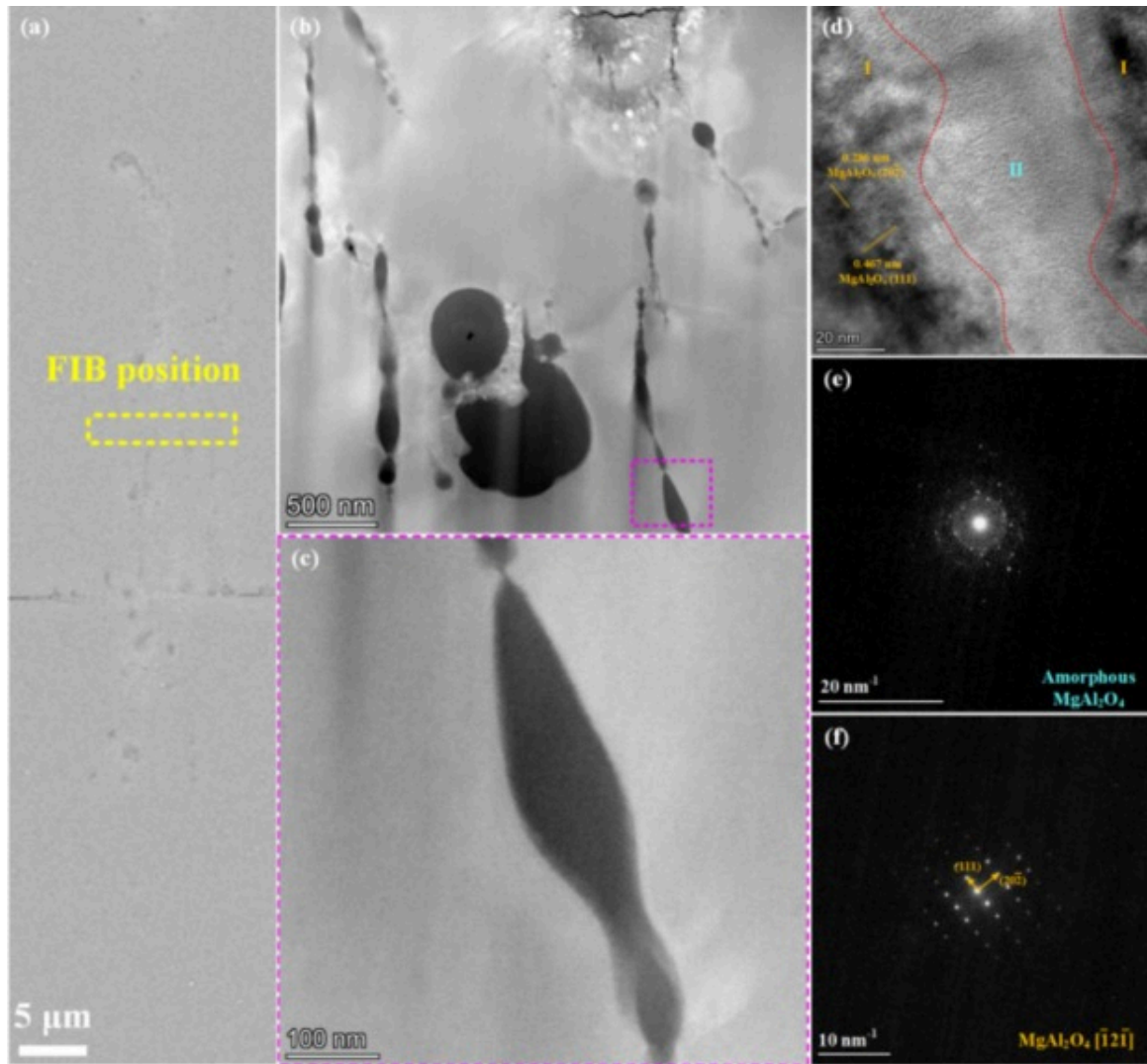
For further analysis of the unidentified region in Fig. 5, TEM observation was adopted and the results are shown in Fig. 6 and Fig. 7. FIB (Focused Ion beam) position was firstly chosen at the interface. As can be seen from Fig. 6b, there are two different phases whose colors are dark gray and light gray respectively, which are distributed in the joint. According to the element mapping analysis in Fig. 6c–e, it can be seen that the elements of O, Mg and Al are evenly distributed in the whole joint with no obvious aggregation, which indicates that there is no difference in element composition between the dark gray phase and the light gray one. By analyzing the diffraction patterns (Fig. 6f–i), it can be demonstrated that aside from MgAl₂O₄ crystal, some ceramics in the welding zone are transformed into amorphous phase after femtosecond laser irradiation. In previous reports, it has also been found nano-scale amorphous phase formed at the interface of femtosecond laser welding ceramic joint [27], but not so large in size. Therefore, another FIB position was chosen at the ceramic side 15 μm away from the interface to identify the phases there and the result was shown in Fig. 7. The experimental results show that some MgAl₂O₄ crystals far from the interface are still transformed into amorphous under the irradiation of femtosecond laser. Moreover, the amorphous structure is extremely unstable and will crystallize rapidly after electron beam irradiation. The amorphous phase formation is surmised as follows. However, after the laser action is over, the melted ceramics suffered a rapid cooling and retain the disordered structure of the liquid state, which leads to the formation of amorphous phase in the re-solidified MgAl₂O₄ ceramics [28]. In the meantime, plasma generated by MgAl₂O₄ transparent ceramics nonlinear absorbing femtosecond laser energy can produce high-pressure conditions during its fast-expanding process, which may also lead to an amorphization of the crystalline [29].



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Fig. 6. (a) SEM image of $\text{MgAl}_2\text{O}_4/\text{MgAl}_2\text{O}_4$ joint; TEM analysis results of $\text{MgAl}_2\text{O}_4/\text{MgAl}_2\text{O}_4$ interface: (b) the HADDF image; (c–e) the element distributions of Mg, Al and O; (f) the HRTEM image of the interface; (g) magnification of region I in (f); (h) the diffraction pattern of region I in (f); (i) the diffraction pattern of region II in (f).

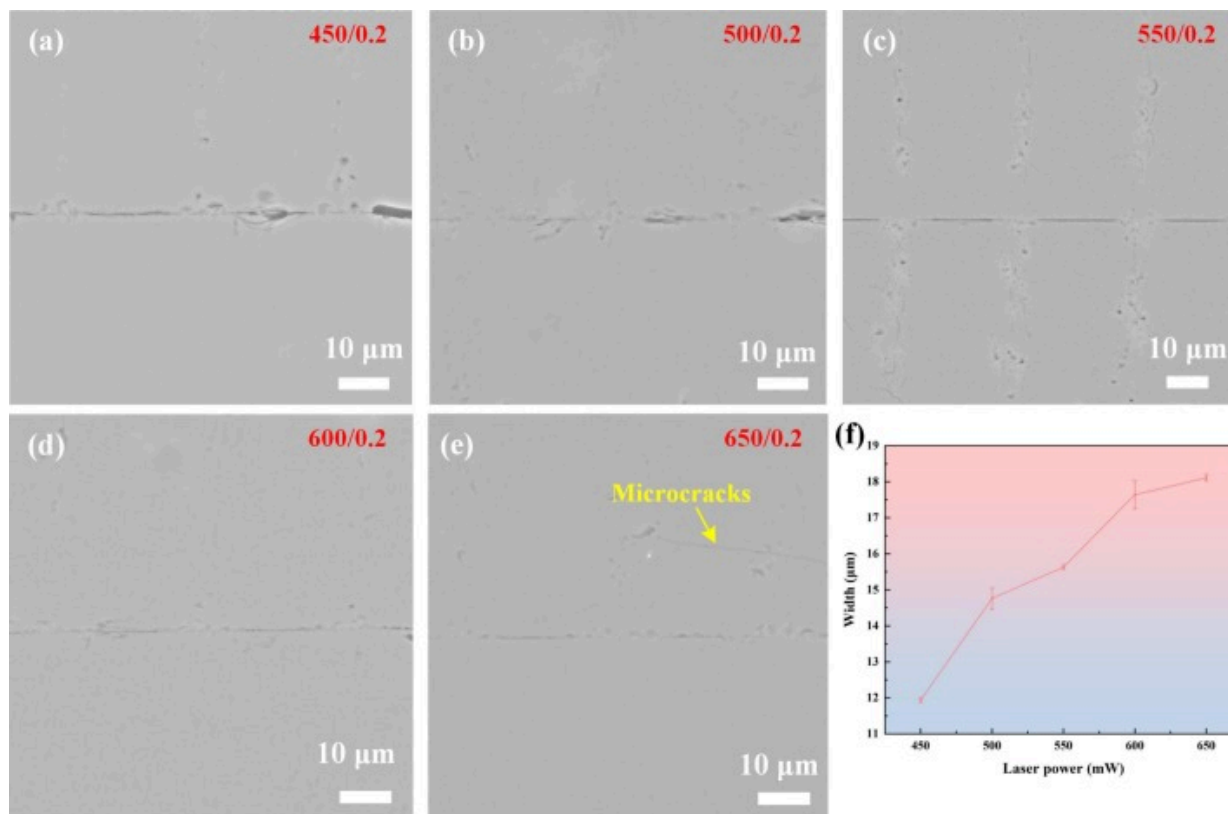


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Fig. 7. (a) SEM image of $\text{MgAl}_2\text{O}_4/\text{MgAl}_2\text{O}_4$ joint; TEM analysis results of MgAl_2O_4 ceramic side: (b) the HADDF image; (c) magnification of (b); (d) the HRTEM image of (c); (e) the diffraction pattern of region II; (f) the diffraction pattern of region I.

Laser power and scanning speed determine the heat input in the welding process [30], which has close relation to the microstructure and mechanical properties of the joint [31,32]. Thus, the effect of the two factors mentioned above were investigated in our research. Firstly, the scanning speed was fixed at 0.2mm/s, and MgAl₂O₄ ceramics are welded by femtosecond laser under different laser powers (450mW, 500mW, 550mW, 600mW, 650mW). As shown in Fig. 8, it can be seen that MgAl₂O₄ ceramics has been bonded together by femtosecond laser at different powers with no obvious defects. The filament-like laser action zone with different contrast can be observed at the interface and there are some black zones in it, which are suspected to be the microvoids left by the melting and sealing of MgAl₂O₄ ceramics under femtosecond laser action. When the laser power is excessively high, microcracks are generated near the interface. With the laser power increases, the width of the welding line gradually increases from ~12μm to ~20μm. The variation of the welding line width shows that the enlargement of femtosecond laser modified area with the increase of the laser power.



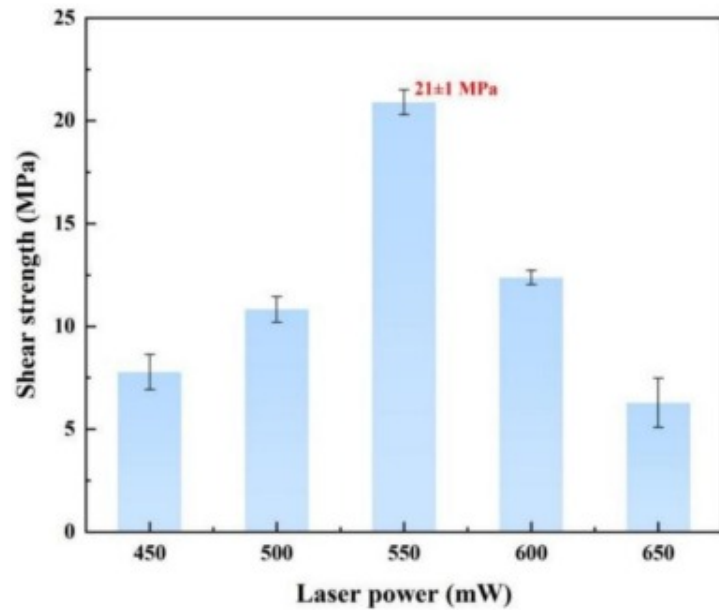
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Fig. 8. The microstructure of MgAl₂O₄/MgAl₂O₄ joints attained under different laser power: (a) 450mW, (b) 500mW, (c) 550mW, (d) 600mW and (e) 650mW; (f) the variation tendency of the welding line width.

The shear strengths of the joints obtained under different power are tested with the result shown in Fig. 9. With the increase of the laser power, the joint strength first increases and then decreases. Combined with the fracture surfaces (Fig. 10), following conclusions can be drawn. When the laser power is low, the heat input to the interface is insufficient caused inadequate joining between the two MgAl₂O₄ ceramics. There is less ceramics left on the welding line indicating that the base materials do not form a good

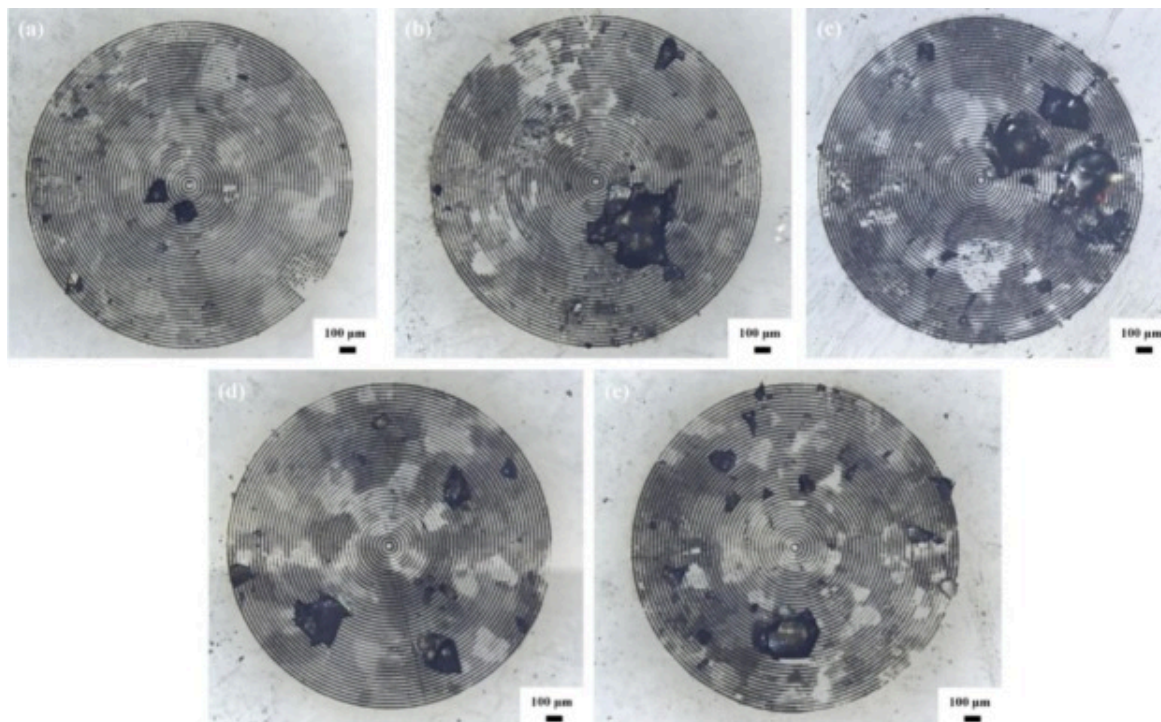
combination, which leads to the low strength of the joint. With the increase of laser power, the increase of effective joining area brings about the uptrend of the shear strength. Anyway, when the laser power is greater than 550mW, the strength of the joint shows a downward trend. Under high power, excessive heat accumulation leads to cracks in ceramics under thermal stress [33], which damages the joint strength.



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Fig. 9. The shear strength of the MgAl₂O₄/MgAl₂O₄ joints attained under different laser power.

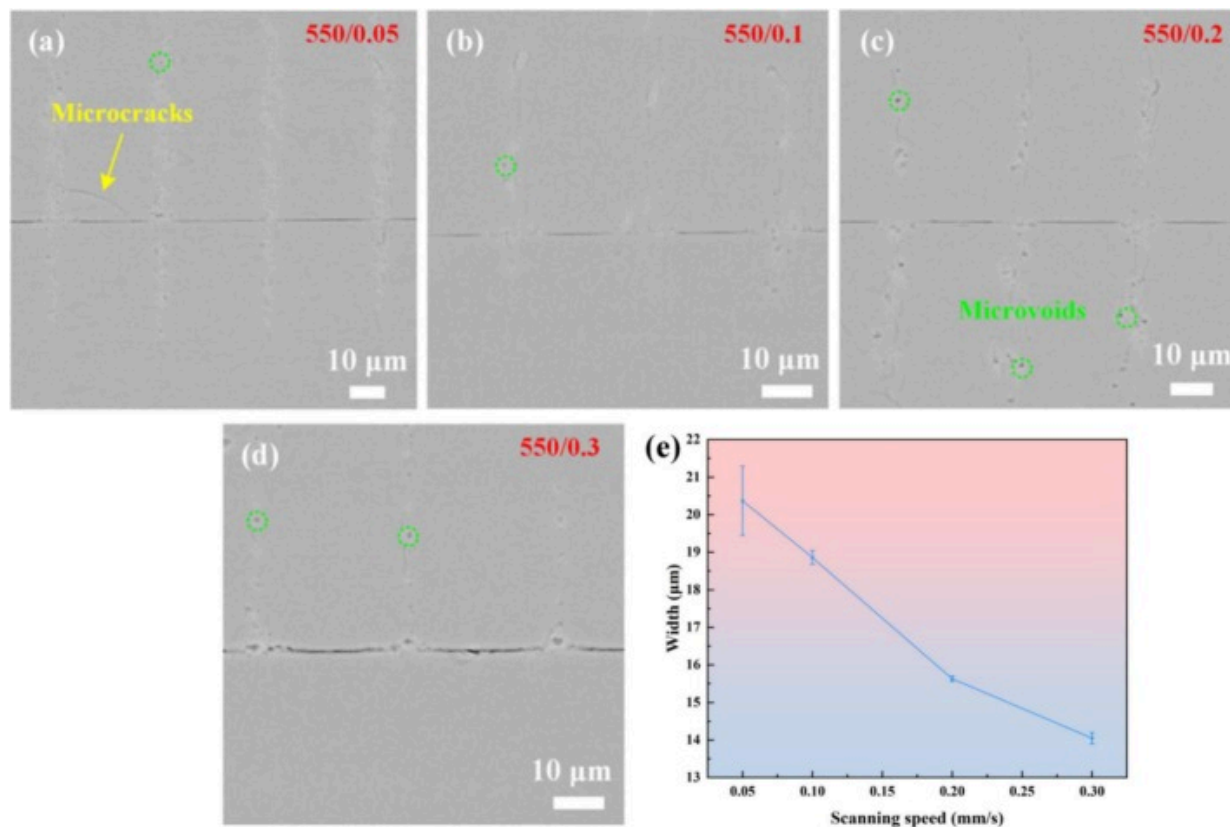


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Fig. 10. The macroscopic morphologies of the joint fracture surfaces attained under different laser power: (a) 450mW, (b) 500mW, (c) 550mW, (d) 600mW and (e) 650mW.

Then the laser power was fixed at 550mW, which is the best parameter in the experiment range, the effect of different scanning speeds is explored. Fig. 11 depicts the microstructure of the joint obtained at the scanning speed of 0.05mm/s, 0.1 mm/s, 0.2mm/s, 0.3mm/s, respectively. Successful bonding between MgAl₂O₄ ceramics was gained without obvious defects. Similarly, filament-like areas can be observed in the joints obtained under different scanning speeds. When the scanning speed is 0.05mm/s, there are several microcracks in the joints. With the rise of scanning speed, the width of welding line decreases from ~20μm to ~14μm.



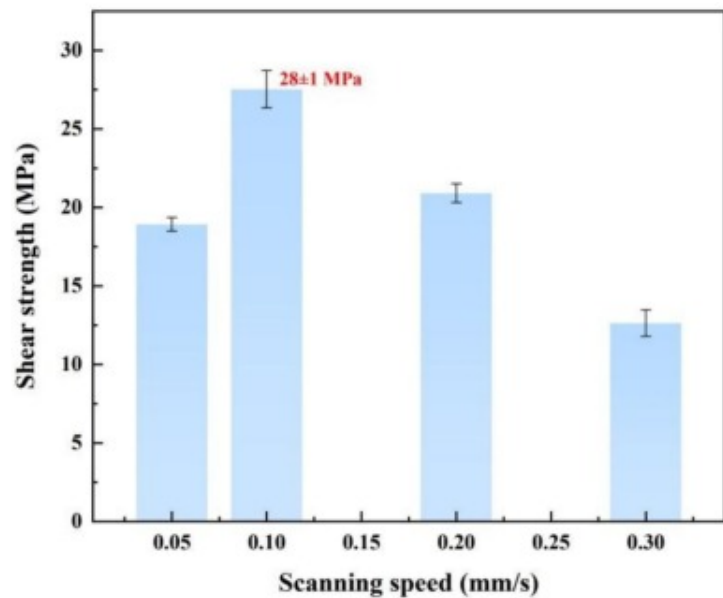
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Fig. 11. The microstructure of MgAl₂O₄/MgAl₂O₄ joints obtained under different scanning speed: (a) 0.05 mm/s, (b) 0.1 mm/s, (c) 0.2 mm/s and (d) 0.3 mm/s; (e) the variation tendency of the welding line width.

As shown in Fig. 12, the joint strength shows the same trend as the laser power change. When the scanning speed is slow, the excessive accumulation of heat increases the risk of cracks and ablation on the ceramic side, which reduces the strength of the joint. With the increase of scanning speed, the moderate energy accumulation at the interface attributes to the formation of a defect-free joint with relatively high strength. Nevertheless, the acceleration of scanning speed induces the decline of joint strength. It is the decrease of

the effective pulse number acting on the unit length that leads to the insufficient melting and joining of ceramics. When the scanning speed is 0.1 mm/s, the joint has the highest strength of 28 MPa.



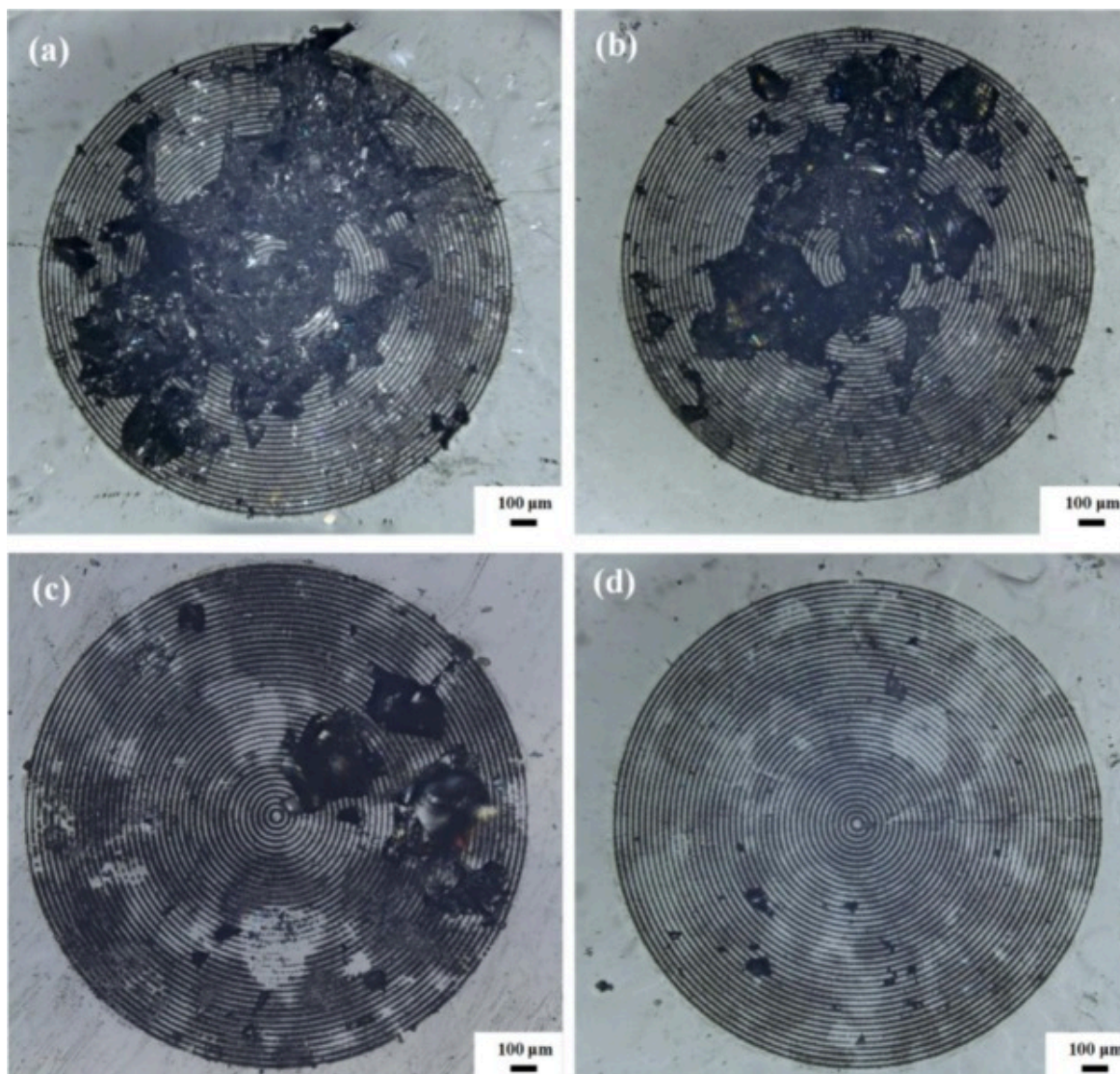
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Fig. 12. The shear strength of the MgAl₂O₄/MgAl₂O₄ joints attained under different scanning speed.

Fig. 13 shows the macroscopic fracture morphologies of the joint obtained at different scanning speeds. As can be seen from Fig. 13, when the scanning speed is 0.05 mm/s, there is a large amount of MgAl₂O₄ ceramics covering many welding lines. This is because strong heat accumulation caused large thermal stress inside the ceramics, which may give rise to the initiation of microcracks. In the process of applying shear force, the MgAl₂O₄ ceramics will rupture by crack propagation, which become the weak part of the joint. When the scanning speed increases to 0.3 mm/s, there is almost no ceramic residue at the welding lines. The higher scanning speed reduces the heat accumulation per unit area,

leading to an untight combination between the two parent MgAl_2O_4 ceramics, which diminishes the joint strength. The crack gradually extends into the ceramic from the interface when the scanning speed set at 0.1 mm/s, and the ceramic remaining on the welding area covers many welding lines, which is consistent with the joint strength.



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Fig. 13. The macroscopic morphologies of the joint fracture surfaces attained under different scanning speed: (a) 0.05 mm/s, (b) 0.1 mm/s, (c) 0.2 mm/s and (d) 0.3 mm/s.

4. Conclusion

In the process of continuous laser or long pulse laser welding transparent materials, light absorption layer always needs to be added to absorb energy to form melting zone. This will not only reduce the original optical properties of the transparent materials after welding, but also lead to thermal damage, cracks and other defects due to its huge heat input [[34], [35], [36]]. However, the appearance of ultrashort pulse laser overcomes the problems mentioned above and shows unique advantages in the field of welding transparent hard and brittle materials. In our research, MgAl₂O₄ ceramics was successfully welded by ultrashort pulse laser thanks to its extremely high peak power density. Under the irradiation of femtosecond laser, parts of MgAl₂O₄ ceramic were modified and transformed into amorphous. Besides, no new phase was generated during the welding process. In the meantime, the effects of laser power and scanning speed on the microstructure and mechanical properties of MgAl₂O₄/MgAl₂O₄ joint were clarified in the general. With the rise of the laser power or the acceleration of the scanning speed, the joint strength first increases and then decreases. After optimization, the best parameter in the experimental range is 550mW/0.1 mm/s, when the highest shear strength is 28MPa. This study offers a new corroboration for the reliable joining of transparent hard and brittle materials by ultrashort pulse laser. By this token, ultrashort pulse laser opens a new door for the effective joining of transparent hard and brittle materials. It can be predicted that ultrashort pulse laser possesses infinite prospect and great potential in the welding field.

CRedit authorship contribution statement

Hao Jiang: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Chun Li:** Writing – review & editing, Project administration, Formal analysis. **Xiaojian Mao:** Methodology, Investigation, Funding acquisition. **Hao Yuan:**

Resources, Data curation, Conceptualization. **Wendi Zhao:** Validation, Supervision, Investigation. **Xiaoqing Si:** Visualization, Methodology, Data curation. **Junlei Qi:** Supervision, Investigation, Formal analysis. **Jian Cao:** Writing – review & editing, Project administration, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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